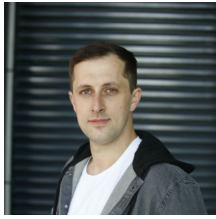


Jan Saro, Ph.D., MBA



Assistant Professor, Czech University of Life Sciences Prague | Junior Researcher, ÚTIA, Czech Academy of Sciences

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Date of birth: 13 May 1990 | **Citizenship:** Czech Republic

Research Profile

I work at the intersection of artificial intelligence, machine learning, operations research, and decision-support systems, with applications in dairy production, sustainable agriculture, and farm-level decision-making. My research emphasises predictive modelling, stochastic / Markov processes, and deployable data-driven tools that translate state-of-the-art AI into something a working farmer can actually use. I combine academic research, software engineering, and quantitative modelling to ship practical AI systems with real-world impact.

Education

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|-----------------------|--|
| 2022 – 2025 | Ph.D. in Systems Engineering
Department of Systems Engineering, Faculty of Economics and Management,
Czech University of Life Sciences Prague.
<i>Thesis:</i> Data-driven decision support systems and predictive modelling in dairy herd health.
<i>Supervisor:</i> Prof. RNDr. Helena Brožová, CSc. |
| Jan – Mar 2025 | Data Science Bootcamp , Le Wagon, London.
Machine learning, deep learning, Python, SQL, neural networks, data engineering. |
| 2021 – 2022 | MBA – Master of Business Administration , Newton University, Prague. |
| 2012 – 2015 | Master's in Open Informatics – Computer Engineering
Czech Technical University in Prague, Faculty of Electrical Engineering.
<i>Thesis:</i> Distributed Task Mapping in Reconfigurable Networked Embedded Systems.
<i>Supervisor:</i> Assoc. Prof. Ing. Přemysl Šůcha, Ph.D. |
| 2009 – 2012 | Bachelor's in Open Informatics – Computer Systems
Czech Technical University in Prague, Faculty of Electrical Engineering.
<i>Thesis:</i> STM32F4xx and Optoelectronic Sensors Interfacing. |

Academic Positions

- 2025 – present** **Assistant Professor**, Department of Systems Engineering, Faculty of Economics and Management, Czech University of Life Sciences Prague (CZU).
- 2025 – present** **Junior Researcher**, Institute of Information Theory and Automation (ÚTIA), Czech Academy of Sciences (AV ČR).

Peer-Reviewed Publications

1. **Saro, J.**; Duchaček, J.; Stádník, L.; Brožová, H. (2026). *Decision support systems in dairy cows farming: a 20-year scoping review of characteristics, applications, and future challenges*. Czech Journal of Animal Science. doi:10.17221/40/2026-CJAS.
2. Bartoňová, K.; **Saro, J.**; Maitah, M.; Duchaček, J. (2025). *Regional patterns and cluster analysis of agricultural methane emissions in the EU-27 countries*. Czech Journal of Animal Science. doi:10.17221/26/2025-CJAS.
3. **Saro, J.**; Šubrt, T.; Brožová, H.; Hlavatý, R.; Rydval, J.; Duchaček, J.; Stádník, L. (2024). *A decision support system for herd health management for dairy farms*. Czech Journal of Animal Science, 69(12), 525–534. doi:10.17221/178/2024-CJAS.
4. **Saro, J.**; Duchaček, J.; Brožová, H.; Stádník, L.; Bláhová, P.; Horáková, T.; Hlavatý, R. (2024). *Discrete Homogeneous and Non-Homogeneous Markov Chains Enhance Predictive Modelling for Dairy Cow Diseases*. Animals, 14(17), 2542. doi:10.3390/ani14172542. *Awarded 1st Prize – Dean’s Award for Best Publication by a Ph.D. Student (FEM CZU, 2024)*.
5. **Saro, J.**; Stádník, L.; Bláhová, P.; Huguet, S.; Brožová, H.; Duchaček, J. (2024). *A decision support system based on disease scoring enables dairy farmers to proactively improve herd health*. Czech Journal of Animal Science, 69(5), 183–192. doi:10.17221/53/2024-CJAS. *Awarded 5th Prize – Dean’s Award for Best Publication by a Ph.D. Student (FEM CZU, 2024)*.

Awards and Honours

- **Finalist**, Falling Walls Lab Czech Republic, 2024 — *Breaking the Wall of the Dairy Farmer AI Assistant*. falling-walls.cz/speakers-2024.
- **1st Prize** – Dean’s Award for the Best Scientific Publication by a Ph.D. Student, FEM CZU, 2024.
- **5th Prize** – Dean’s Award for Best Publication by a Ph.D. Student, FEM CZU, 2024.

Industry and Applied Experience

- Jan – Jun 2023** **Web Developer**, Avast / Gen Digital. Development of web applications for security and privacy services.
- 2019 – 2023** **Developer**, ISFG (start-up). Information systems for the Czech public sector.

2018 – 2019	Full-Stack Web Developer , S&T CZ s.r.o. Development of the EESSI cross-border social-security data exchange system (.NET Core, Java EE, ReactJS, PostgreSQL, Azure DevOps).
2015 – 2018	Software Developer & Consultant , NextSoft spol. s r.o. Payment-card systems (HP-Unix) and internal web applications.
Aug – Dec 2013	Internship – BPM Developer , Center of Knowledge Management, Czech Technical University. Implementation of BPMN processes.
Jul – Aug 2012	Internship – Java SE Developer , ELVAC PRAHA s.r.o. Desktop visualisation tool for industrial safety systems.

Teaching, Supervision and Public Engagement

- Master's-level instruction in Systems Engineering and Decision Support, Faculty of Economics and Management, CULS Prague (2025 – present).
- Falling Walls Lab Czech Republic 2024 finalist: a public-facing talk on dairy-farmer AI assistance, deliberately framed for non-specialist audiences.
- Freelance web-development practice (> 50 delivered websites) used for hands-on teaching examples of real-world software constraints.

Technical Skills

Modelling & ML	Python, scikit-learn, gradient boosting (XGBoost, LightGBM), Markov chains, mixed-effects models, agent-based simulation, MATLAB.
Programming	.NET Core, Java, C#, C, PHP, SQL, JavaScript, HTML5, CSS3, Assembler, VHDL, Verilog.
Frameworks	Entity Framework, Nette, Angular, ReactJS, Bootstrap, WordPress, jQuery.
Databases	MySQL, MSSQL, PostgreSQL, Oracle.
Tools	Git, Azure DevOps, IBM Cognos, Power Apps.
Operating systems	Windows, Unix (HP-Unix).
Embedded systems	STM32, Arduino.

Languages

- **Czech** — native speaker.
- **English** — C1 (Common European Framework).

Courses and Certificates

- Data Science Bootcamp, Le Wagon, London (2025).

- Kaplan International, Manchester – Upper Intermediate English (2019).
- CCNA Exploration: Network Fundamentals – Cisco (2014).
- British Study Centres, Oxford – Intermediate English (2013).
- CELT Cardiff – Intermediate English (2011).
- Electrical Qualification §5 (Decree 50/1978 Coll., Czech Republic).
- Category B driving licence.

References

Prof. RNDr. Helena Brožová, CSc.

Ph.D. supervisor

Department of Systems Engineering, FEM,
CULS Prague

brozova@pef.czu.cz

Assoc. Prof. Tatiana Valentine Guy, Ph.D.

Head of the Department of Adaptive Systems

Institute of Information Theory and Automation
(ÚTIA), Czech Academy of Sciences

guy@utia.cas.cz

Additional references and reprints available on request.